

CARBENTRA / CARBENTRA–P16–EVT–B INTEGRATED ELECTRICAL SOURCE

101 main–board parts + 4 remote–head parts + 4 off–board assemblies; harness models are not PCB parts

Hierarchy is functional grouping. Shared global net names are real electrical connections.

CANDIDATE ONLY: no fabrication, energization, safety or certification release

1 controller

PS101 IRM–10–5; AP63203 3.3 V rail
ESP32–C3 + low–side relay driver
K101: HOT_LOAD to HOT_SWITCHED
Coil feed = hardware–gated COIL_5V
OUTPUT_PRESENT_N enters U101.15

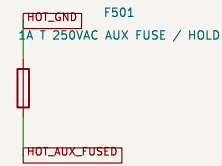
File: integrated_controller.kicad_sch

2 meter

RS201 Kelvin shunt + voltage divider
ATM90E26 AFE + reinforced SPI isolation
R05CT isolated supply for HOT_3V3
Powered ASV clock / CX201 shown at right
HOT_GND is live line reference

File: integrated_meter.kicad_sch

AUXILIARY FUSE: protected line to isolated supply input



3 feedback

Four 33k resistors + ACPL–K376 detector
Senses HOT_SWITCHED relative to HOT_N
OUTPUT_PRESENT_N returns to the MCU
Active–low 100/120 Hz pulses
No proof of dead output or contact position

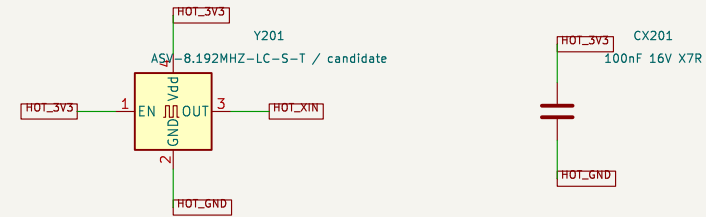
File: integrated_feedback.kicad_sch

4 thermal

Remote head: U401 / R401 / C401 / J404
TMP302 + supervisor + trip–dominant latch
SW401 physical rearm only; no MCU rearm
Q401 gates coil feed; TP401 / TP402 optional
Backside coupling / insulation needs qualification

File: integrated_thermal.kicad_sch

POWERED CLOCK: OSCO is NC; exact frequency procurement HOLD



5 assembly protection

Plug L –> F601 main fuse –> TF601 thermal link –> protected main PCB input
Socket output follows K101; PE is direct and never enters PCB or relay

File: integrated_assembly.kicad_sch

#FLG1
EXTERNAL / PASSIVE RAIL



#FLG5
EXTERNAL / PASSIVE RAIL



#FLG2
EXTERNAL / PASSIVE RAIL



#FLG6
EXTERNAL / PASSIVE RAIL



#FLG3
EXTERNAL / PASSIVE RAIL



#FLG7
NC LIVE–POTENTIAL NET ANCHOR



#FLG4
EXTERNAL / PASSIVE RAIL



HOT_GND is LINE POTENTIAL. PE is external, continuous and never switched.

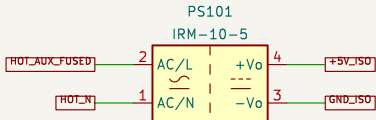
Remote head: component face rear; smooth GND copper backside to qualified insulating pad. Thermal lag/insulation HOLD.

Sheet: /		
File: integrated.kicad_sch		
Title: CARBENTRA CARBENTRA–P16–EVT–B integrated source		
Size: A3	Date:	Rev: EVT–B HOLD
KiCad E.D.A. 9.0.2		Id: 6/6

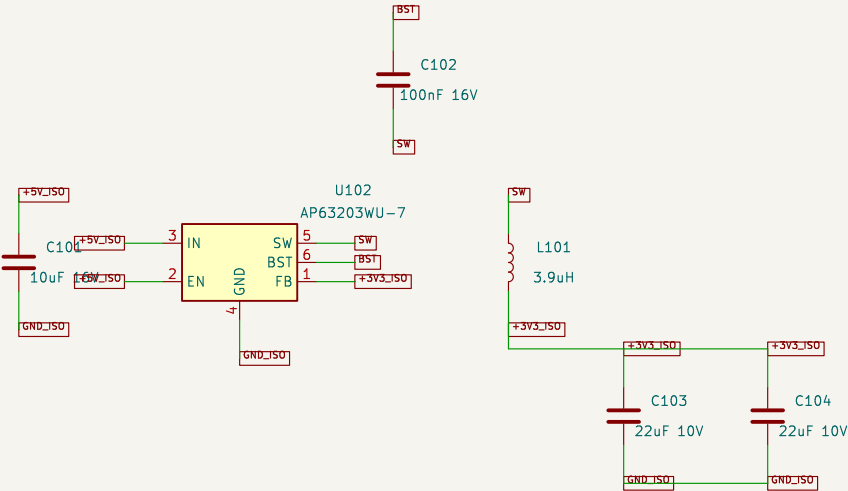
Off-board coordinated branch fuse + thermal cutoff. F501 is the on-board auxiliary fuse. PE remains continuous and unswitched.

SPI metering, output feedback and thermal permission are directly integrated; no inter-module host connector.

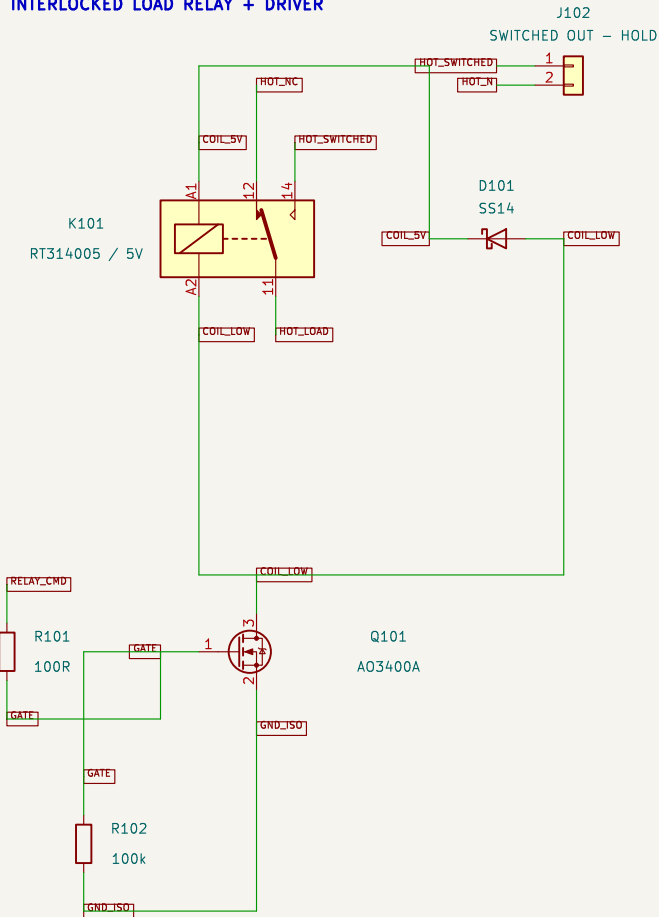
01 ISOLATED INPUT



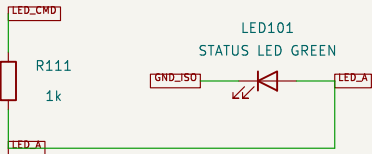
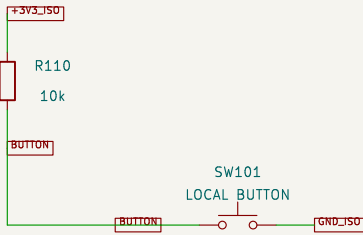
02 BUCK CONVERTER



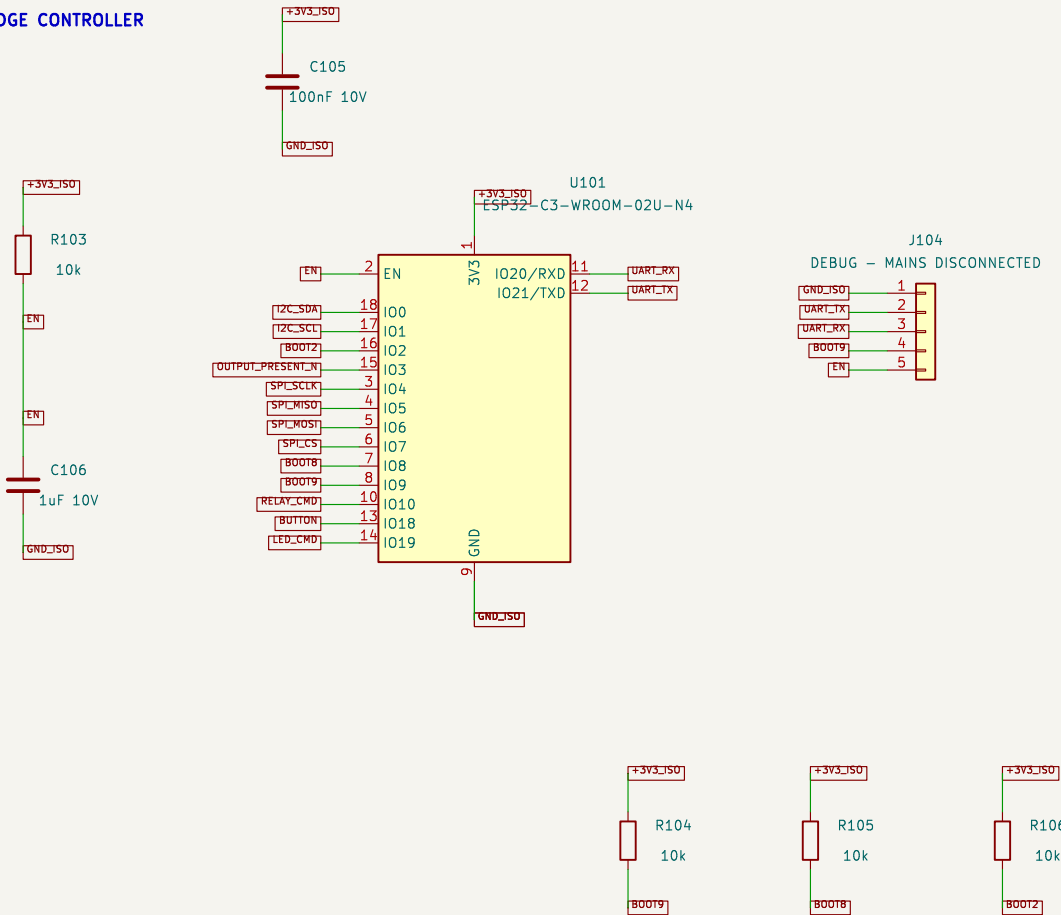
03 INTERLOCKED LOAD RELAY + DRIVER



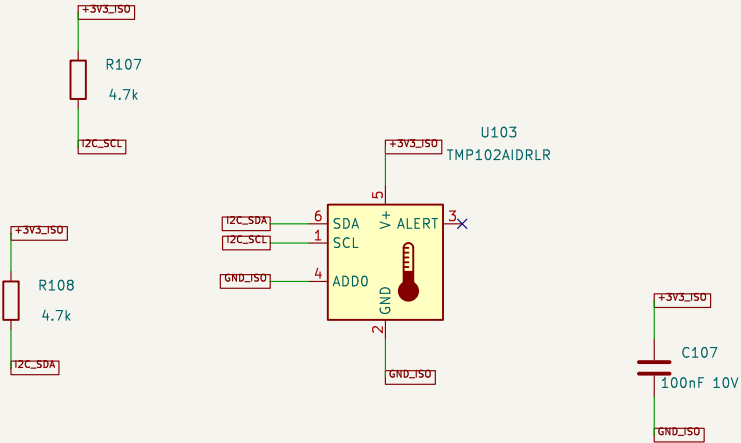
05 LOCAL CONTROLS



04 EDGE CONTROLLER



06 BOARD TEMPERATURE



HOT_* = LIVE LINE REFERENCED. Never connect hot ground to PE, USB, isolated ground or accessible metal.

Protected line -> Kelvin shunt -> relay COM. Voltage samples input L-N; separate ACPL-K376 senses switched output.

01 LOAD SHUNT + MATCHED RC

03 ATM90E26 AFE

04 REINFORCED SPI BARRIER

02 VOLTAGE DIVIDER + FILTER

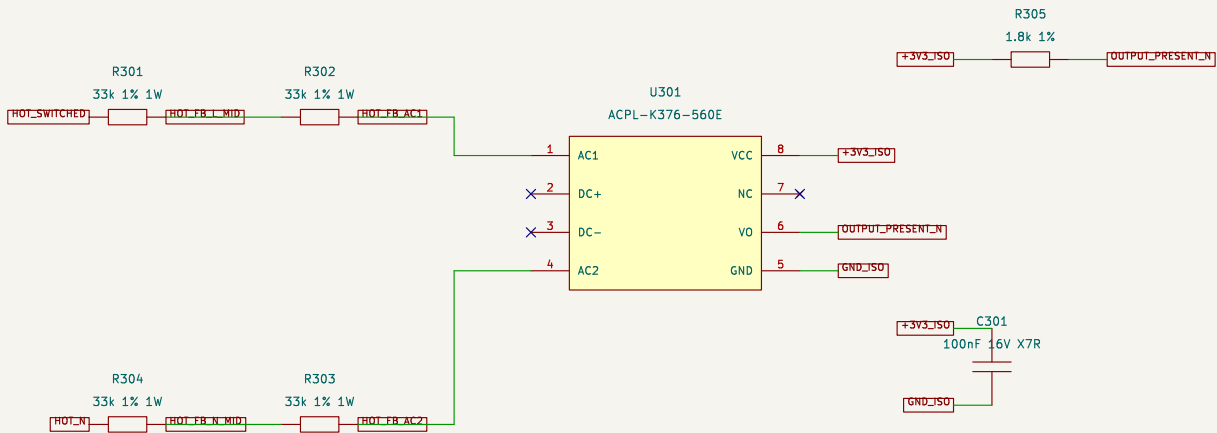
05 ISOLATED POWER 5V -> 3.3V

06 RESET / DECOUPLING; POWERED CLOCK ON OVERVIEW

REV B OUTPUT PRESENCE / DEVELOPMENT ONLY

198–242 VAC OPERATING PROPOSAL; 264 VAC ANALYSIS HEADROOM ONLY; QUALIFICATION HOLD

MAINS INPUT: tap the ACTUAL socket contacts after all switched poles



ISOLATED 3.3 V: active–low pulses, 100/120 Hz when AC is present

- DC+ / DC– / NC intentionally open. Do not bridge either isolation domain.
- No safety isolation proof, no proof of absence of voltage, no physical contact–position feedback.
- 132k total series resistance; built–in bridge/threshold controller inside U1.
- Use a separate hardware thermal interlock AND a qualified series thermal cutoff.

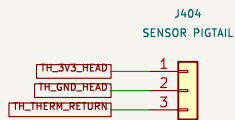
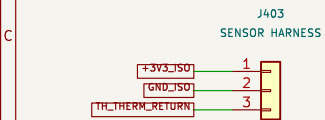
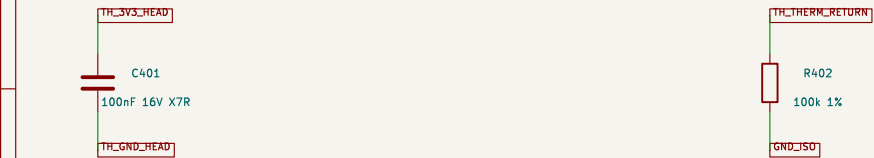
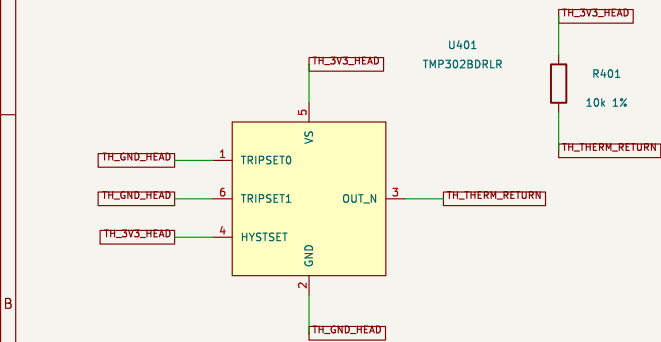
Sheet: /3 feedback/ File: integrated_feedback.kicad_sch		
Title: CARBENTRA EVT–B: feedback		
Size: A4	Date:	Rev: EVT–B HOLD
KiCad E.D.A. 9.0.2		Id: 4/6

REV B HARDWARE THERMAL TRIP + LOCAL PHYSICAL REARM ONLY

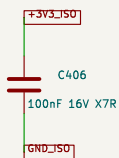
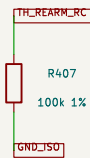
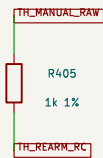
DEVELOPMENT CANDIDATE – NO ENERGIZATION OR FABRICATION RELEASE

REMOTE HEAD: U401 / R401 / C401 / J404 only. W401/W402 are cable models, not PCB parts.

01 REMOTE TEMPERATURE HEAD



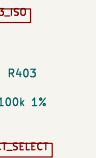
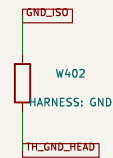
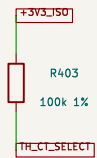
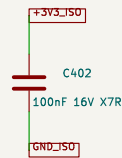
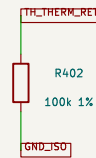
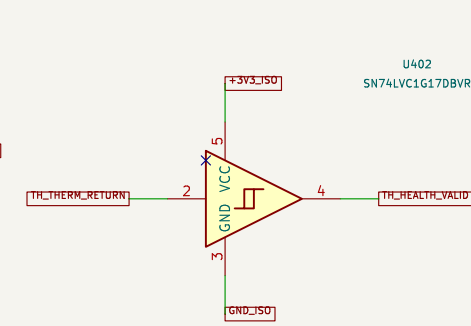
05 FRESH REARM EDGE (DO NOT GATE CLOCK WITH READY)



COIL_5V feeds K101.A1 and D101 cathode; no ungated coil-supply bypass is permitted.

A coil interlock cannot interrupt welded load contacts. Separate qualified line-series thermal cutoff remains required.

02 WIRE-OPEN FAIL-LOW



R10 ASSEMBLY PROTECTION / OFF-BOARD INTERNAL PARTS

CANDIDATE ONLY: this sheet closes the drawn assembly path, not qualification or certification

A

B

C

D

E

F

A

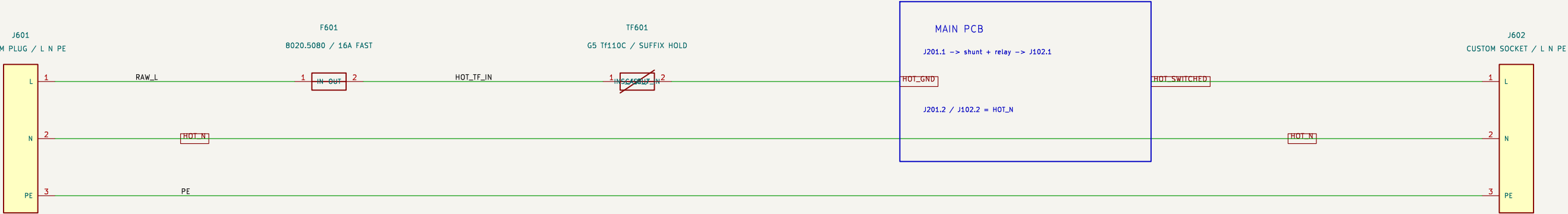
B

C

D

E

F



PE: direct plug-to-socket path; NEVER through PCB, relay, fuse or software

F601 = MAIN 16 A fast-acting fuse. F501 = separate PCB auxiliary 1 A time-delay fuse.

TF601 = distinct one-shot MICROTEMP G5 thermal link, provisional Tf110C; order suffix and mounting HOLD.

SHF16A:1.5kA@250VAC cos(phi)0.7-0.8. Typical melting I2t760A2s is NOT total clearing or a downstream energy bound.

Upstream breaker, prospective fault current, fuse I2t/selectivity, hot-condition derating and wiring coordination are UNVERIFIED.

NO ACTIVE CURRENT LIMIT: a fuse rating is not a continuous-current clamp or a guaranteed socket overload threshold.

Custom contacts + 2x CQP8040.0003 clips;45x16x2 carrier / Cu collectors; fit accepted, heating/insulation HOLD.

Assembly contact numbers identify L/N/PE wiring. SHF8020.5080 fuse6.3x32mm replaces216/OGN for higher clip margin.

The main-PCB box references real circuits on sheets 1-4; it is not an electrical bypass between HOT_GND and HOT_SWITCHED.

PE continuity, polarity, touch protection, contact heating and cold-weld disconnection still require physical verification.

F601 / TF601 / J601 / J602 are on_board=false. Main PCB part count remains 101.